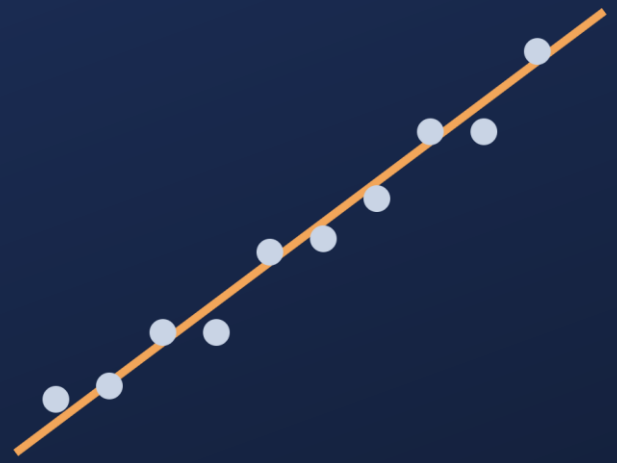
ML & AI FOR LIFE SCIENCES 2026 · DAY 1 · LECTURE 3

Supervised learning: linear regression

Predicting a number, and checking the prediction honestly



Trần Thượng Vân Du · SIB Swiss Institute of Bioinformatics

VNUHCM-University of Science · 02/11/2026

YOU ALREADY KNOW THIS

VI: giải thích hay dự đoán?

Same equation, different question

Statistics: explain

Is BMI associated with blood pressure?

Look at: coefficient, confidence interval, **p-value** on the data you have.

Machine learning: predict

How well can I predict blood pressure of a **new** patient?

Look at: **error on unseen data**. Coefficients matter less.

A model can be significant and still predict badly, or predict well and explain nothing.

SUPERVISED LEARNING

VI: hồi quy · phân loại

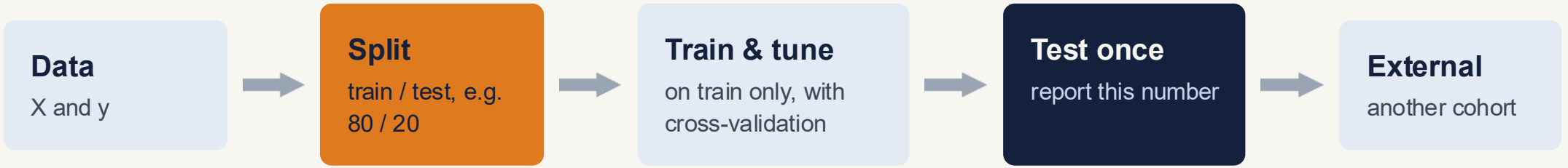
Learn f so that f(X) ≈ y, from labelled examples

	Regression	Classification
y is…	a number	a category
Question	How much? How many?	Which class? Yes or no?
Example	Days in hospital	Tumour vs normal
Simple model	Linear regression (now)	Logistic regression
Error measured by	RMSE, R^2	Accuracy, AUC, F1

THE WORKFLOW

VI: tập huấn luyện · tập kiểm tra

Split first; touch the test set only once, at the end



The test set plays the role of **future patients**. If it influenced any choice, it no longer can.

Many biological questions predict a number

Age from DNA methylation

CpG sites → chronological age ("epigenetic clock")

Drug response

Cell-line expression → IC50 of a drug

Gene expression

Promoter sequence → expression level

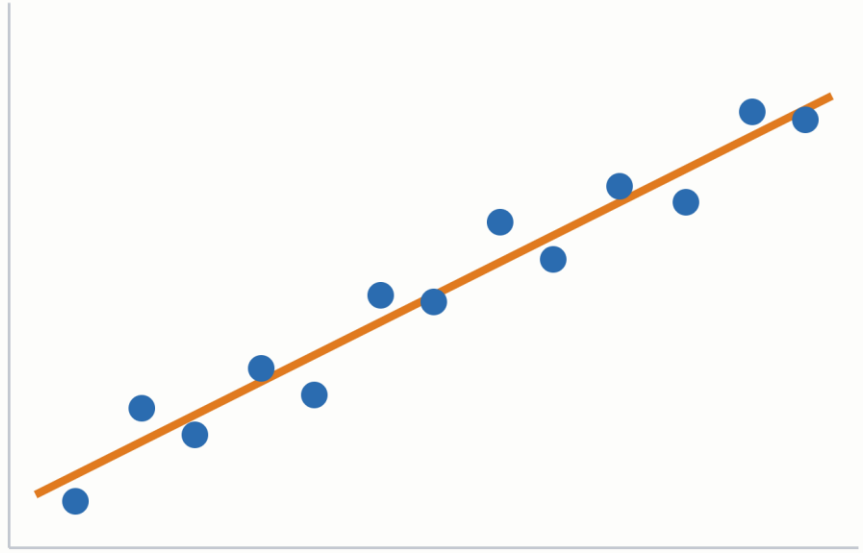
Clinical markers

Metabolites → future HbA1c or eGFR

THE MODEL

VI: hệ số chặn · độ dốc

A straight line with two numbers to learn



$$\hat{y} = \theta_0 + \theta_1 \cdot x$$

θ_0 intercept: prediction when x = 0

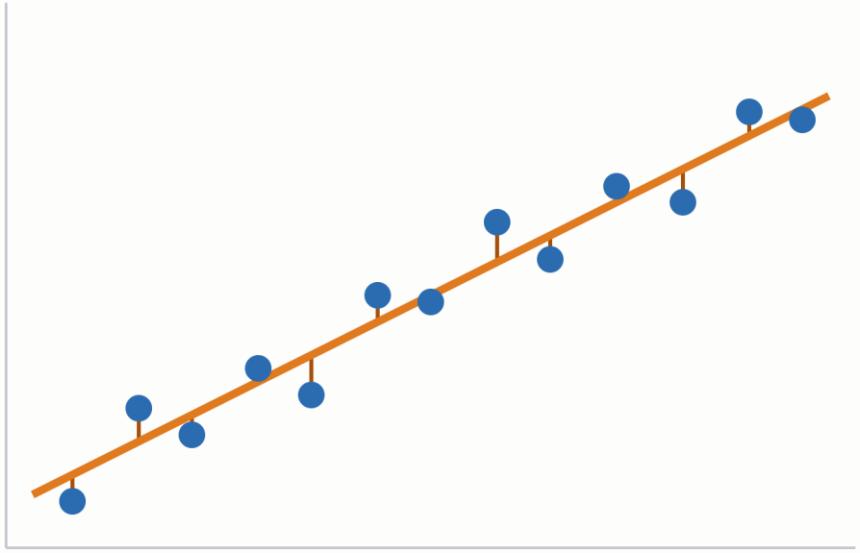
θ_1 slope: change in ŷ per unit of x

Learning = finding the θ values that make the line fit best. But what is "best"?

THE LOSS FUNCTION

VI: phần dư · hàm mất mát

"Best" = smallest average squared error



Residual = observed − predicted (the short vertical lines).

$$MSE = mean((y - \hat{y})^2)$$

Squaring makes all errors positive and punishes big misses more.

"Ordinary least squares" = the line with minimum MSE.

VI: hạ gradient · tốc độ học

Gradient descent: walk downhill on the loss



loss vs parameter θ_1

1 · Start with random θ.

2 · Compute the slope (gradient) of the loss.

3 · Step downhill; repeat.

Learning rate = step size. Too big: overshoot. Too small: very slow.

Linear regression also has a direct formula; neural networks (Day 2) rely on this walk.

MULTIPLE REGRESSION

VI: hồi quy đa biến

One weight per feature, read "with the others held fixed"

$$\hat{y} = \theta_0 + \theta_1 \cdot x_1 + \theta_2 \cdot x_2 + \dots + \theta_p \cdot x_p$$

Example

$$SBP = 90 + 0.5 \cdot age + 1.2 \cdot BMI$$

+1 BMI unit → +1.2 mmHg, at the same age.

Comparing weights

Size of θ depends on units. Standardise features first to compare them.

Correlated features

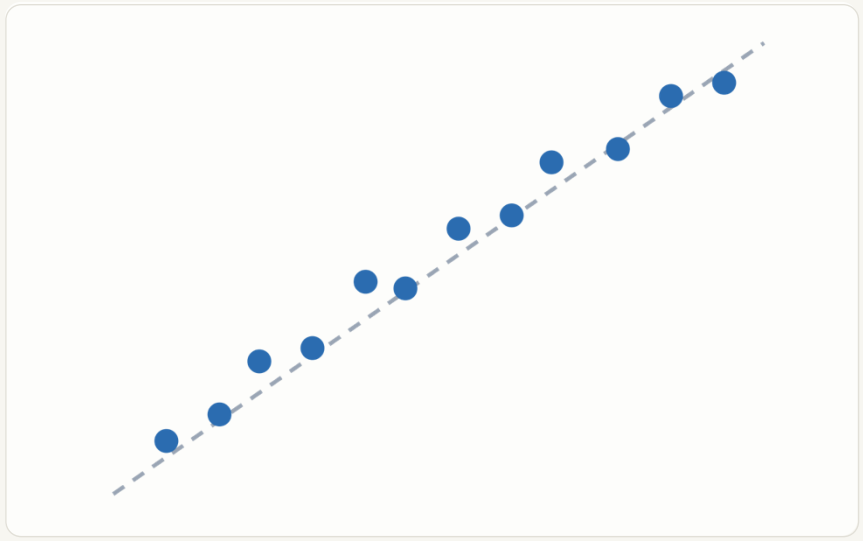
Two co-expressed genes share the credit; weights become unstable and can flip sign.

EVALUATION · LOOK FIRST

VI: đánh giá mô hình

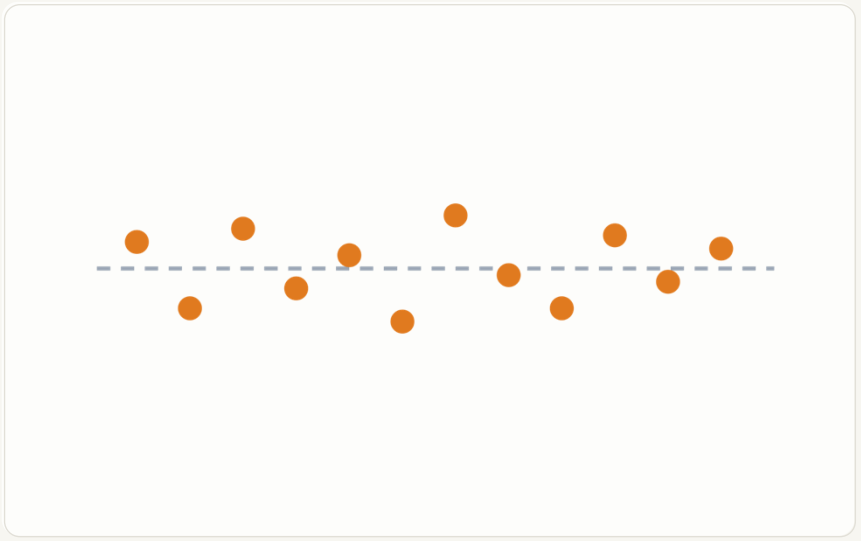
Two plots tell you more than one number

Observed vs predicted



Good model: points hug the dashed diagonal.

Residuals vs predicted



Good: a shapeless cloud around 0. A curve or funnel = the model misses structure.

EVALUATION · NUMBERS

VI: sai số · hệ số xác định

Three metrics for regression

RMSE

$$\sqrt{mean((y - \hat{y})^2)}$$

Typical error, in the units of y.

MAE

$$mean(|y - \hat{y}|)$$

Same units; less sensitive to a few large misses.

R²

$$1 - SSres / SStot$$

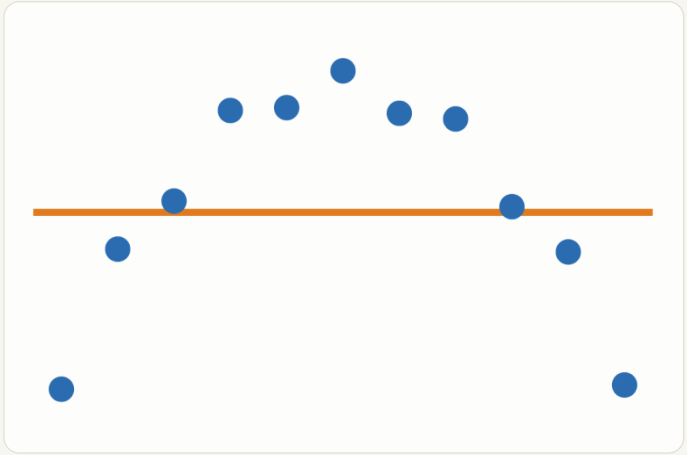
Share of variance explained vs predicting the mean. 1 = perfect; 0 = no better than the mean; can be negative on test data.

Always report them on test data, next to a baseline (predict the training mean).

GENERALISATION

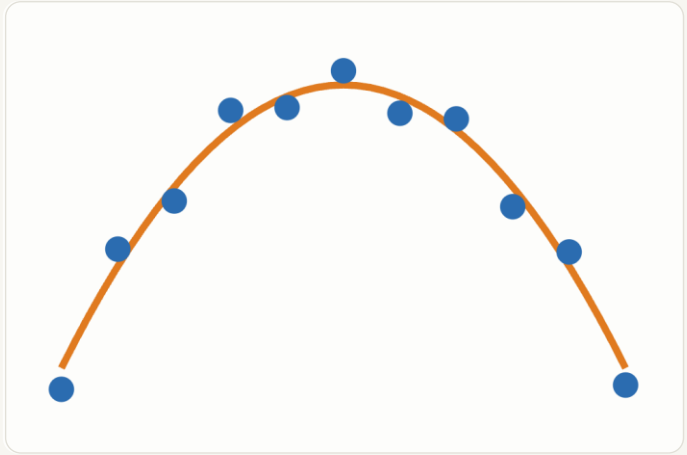
VI: thiếu khớp · quá khớp

Too simple, just right, too flexible



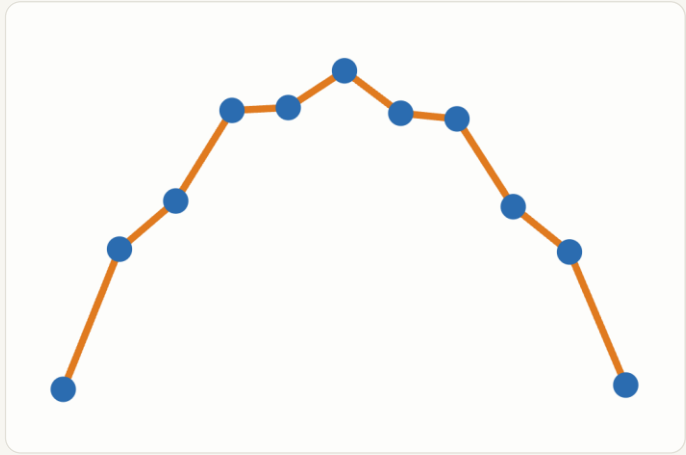
Underfit

High error on train and test (high bias)



Good fit

Captures the trend, ignores the noise



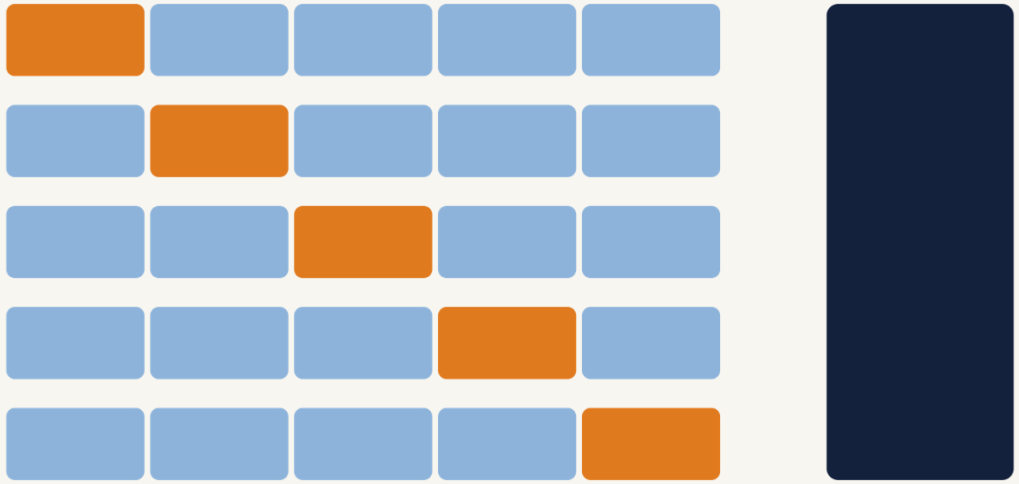
Overfit

Zero train error, poor test error (high variance)

CROSS-VALIDATION

VI: kiểm định chéo k-fold

k-fold cross-validation: every sample is tested once



Light blue = train · **orange** = validation fold · **dark** = test set, kept aside

Train 5 times, average the 5 validation scores → a stable estimate, plus its spread.

Use CV to **choose** settings; use the test set to **report**.

Grouped data: GroupKFold (all samples of a patient in one fold).

THE OMICS PROBLEM

VI: số đặc trưng ≫ số mẫu

When p ≥ n, a linear model fits the training data perfectly

$R^2 = 1.00$

on training data

$R^2 < 0$

on test data

100 patients, 20,000 genes: there are infinitely many perfect lines through the training data.

Most of them are noise.

We need to tell the model: **prefer simple solutions**.

REGULARISATION

VI: chính quy hóa

Add a penalty for large weights

```
minimise MSE + λ · penalty(θ)
```

Model	Penalty	Effect	Use when
Ridge	Σ θ² (L2)	Shrinks all weights	Many small effects
Lasso	Σ |θ| (L1)	Sets many weights to 0	You want a short gene list
Elastic net	L1 + L2	Sparse, keeps correlated groups	Omics default

λ (alpha in scikit-learn) sets the strength: chosen by cross-validation. Standardise features first.

CASE STUDY

VI: đồng hồ biểu sinh

The Horvath epigenetic clock is an elastic net

~8,000

samples, 51 tissues and cell types

353

CpG sites kept by the penalty, out of ~21,000 candidates

3.6 yrs

median absolute error on test data

Same recipe as today: methylation matrix X, age y, elastic net, λ by cross-validation, evaluation on held-out samples. The gap "predicted − real age" became a biomarker in its own right.

Horvath, Genome Biology 2013

If the test set helped choose anything, the score is a lie

Wrong order

1. Select top 100 genes on **all** samples
2. Scale using **all** samples
3. Split train / test
4. Train, test → great score

Works even on random data.

Right order

1. Split train / test first
2. Select genes and fit scaler on **train** only
3. Apply the same steps to test
4. Test once → honest score

scikit-learn Pipeline does this for you, also inside CV.

STRENGTHS AND WEAKNESSES

VI: ưu điểm · nhược điểm

Linear regression: always your first baseline

Strengths

- Fast, simple, reproducible
- Coefficients are interpretable
- With regularisation, works when $p \gg n$
- Hard to beat on small datasets

Weaknesses

- Only straight-line, additive effects
- Interactions must be added by hand
- Sensitive to outliers
- Unstable with correlated features

PITFALLS IN BIOLOGY

VI: sai lầm thường gặp

Four ways a regression model fools you

Extrapolation

Trained on adults 30–70; applied to children.

Confounding

The model predicts age from batch because old samples were run first.

Repeated samples

Visits of one patient in both train and test.

Skewed outcome

Triglycerides, counts, costs: log-transform y first.

TAKE-AWAYS → COLAB

Three things to remember

1

Judge a model by its error on data it never saw: CV to tune, test to report.

2

$p \gg n$ needs regularisation: Ridge, Lasso, Elastic net.

3

Split before any selection or scaling. Use a Pipeline.